

Particle counter calibration

12-May-2010



Jason Kelly, calibration services engineer, Kenelec Scientific Australia asks: is ISO-21501 the complete solution?

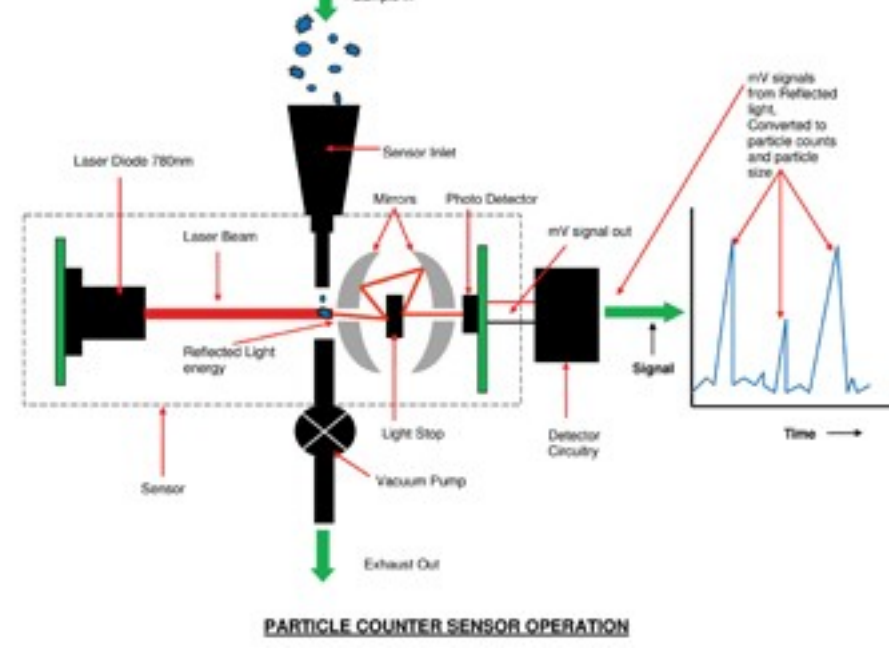


Figure 1: General principle of how a particle counter works Image courtesy of Particle Measuring Systems, Kenelec Scientific Australia & TSI

Jason Kelly, calibration services engineer, Kenelec Scientific Australia asks: is ISO-21501 the complete solution?

With the trend in many industries for cleaner product processing environments, optical particle counters have become increasingly more strategic in environmental monitoring of cleanrooms.

With new cGMP guidelines and traceability, particle counter accuracy is essential in maintaining a quality environmental monitoring system. Therefore with such a big role to play in product quality and ensuring overall product safety, especially in the pharmaceutical, biotech and medical device industries, particle counter accuracy is critical.

To check that particle counters remain accurate once they are calibrated outside of the factory requires some kind of comparison against references, following a well-documented and validated calibration procedure – hence the arrival of ISO 21501.

Calibration – the process of establishing the relationship between a measuring device and the units of measure – is achieved by comparing a device or the output of a device to a standard having known measurement characteristics. To improve the quality of the calibration and have the results accepted by outside organisations, it is desirable for the calibration and subsequent measurements to be “traceable” to internationally defined measurement units. Establishing traceability is accomplished by a formal comparison to a standard that is directly or indirectly related to national standards, international standards, or certified reference materials.

Quality management systems call for an effective metrology system that includes formal, periodic, and documented calibration of all measuring instruments. Thus, ISO 9000 and ISO 17025 require effective calibration systems.

Prior to calibration, it is worth developing an understanding of how particle counters work. The counter takes a sample and the particles in that sample pass through a laser beam thus scattering the light energy, and the electronics inside the sensor convert this light energy into a voltage (see Fig. 1). The voltage is proportionate to the size of the particle or the amount of light the particle scatters. The electronic circuitry picks up the voltage signal from the photo detector (which converted the light energy into the voltage signal). Digital threshold circuitry then sizes and counts the particles.

Primary particle counter calibration is carried out by the manufacturer and is usually conducted following manufacturer procedures and guidelines from the American Society of Testing Materials (ASTM) and Japanese Industrial Standard (JIS) B 9921 where a Condensation Particle Counter (CPC) is used as a reference unit. With this standard the accuracy of the Particle Counter is set (See Fig. 2). Size calibration is checked using a Pulse Height Analyser (PHA).¹

For field calibrations, the previous methods followed were not completely adequate. Particle Counter Service centres across the world used different methods and followed different guidelines.

Particle counter manufacturers went in two different directions with field calibrations. Some chose a calibration method using a calibrated reference unit and followed guidelines in ASTM F649-1 while other particle counter manufacturers used a PHA and followed guidelines in ASTM, F328-98 and some parts of JIS B 9921.

Count efficiency was verified during the factory calibration as well as size calibration using certified reference materials such as polystyrene latex spheres (PSL).

Following the reference particle counter method (also known as count match calibration) the particle counter is calibrated using reference particles and is compared to a calibrated reference unit. This method calibrates the accuracy of the particle counter. Sizing is also calibrated but is based on a channel splitting method and the particle counter is effectively used to calibrate itself (see Fig. 3).

With the PHA method each particle size is calibrated using reference particles (see Fig. 4).

Older particle counters had set electronic thresholds for each particle size. So when a specific channel was calibrated the reference particle size for that channel was sampled through the particle counter, the PHA distribution was displayed and the median voltage was checked against the threshold (see Fig. 5). The new threshold was then set. Other manufacturers had set thresholds for specific sizes, which never changed. The PHA method only calibrated based on size. Therefore a perfectly calibrated particle counter with all channel thresholds aligned to the reference particle sizes could be out of specification in relation to accuracy. Since there was no procedure to check the accuracy, it was taken for granted it should be fine once size was verified and that the reference particles sizes were always the same size from batch to batch (which is not the case as there are slight changes in size).

With so much emphasis on particle counter importance and the various different guidelines, ISO 21501 has been developed to harmonise the current situation. ISO 21501 is at last a complete solution to the calibration of particle counters. It addresses inherent issues within previous particle counter calibration guidelines where either sizing or accuracy calibrations were performed in the field.

First, it requires that both methods be followed (i.e. the PHA for sizing calibration and the Ref Unit for accuracy calibration). Second, Size Resolution must be reported. This is the measure of the ability of the particle counter to distinguish between particles of different size and is the verification of size calibration.

Third, and most important, Count Efficiency must be reported. This is the ratio of counts between the reference unit and the particle counter. The reference unit must be more sensitive than the particle counter with its counting efficiency being a known value, and it needs to be calibrated by a higher sensitive instrument, such as a CPC. Counting efficiency is the verification of the accuracy of the calibration. Figure 6 is an example report that uses 0.5µm reference particles.

ISO 21501 states that counting efficiency should be between 90-110% for particles 1.5 to 2 times greater than the smallest channel and for the smallest channel to have a counting efficiency between 30-70%.

Measurement uncertainty

Although not a requirement of ISO 21501 but a fundamental requirement of ISO 17025: General requirements for the competence of testing and calibration laboratories, the Uncertainty of Measurement (UoM) statement gives the end user confidence in instrument performance. UoM is a complex science and an explanation of the concept can be found in the Guide to the Expression of Uncertainty in Measurement, also known as the ISO GUM. UoM is the metrologist’s nirvana and is widely used in the world of metrology. However, with the right methodology and attitude, UoM can be applied to particle counters with reasonable results, e.g. with a minimum uncertainty of less than 10%. Liquid particle counters should have a lower minimum uncertainty than airborne particle counters.

With the spotlight firmly on particle counters, more customers are calling for recognised accredited quality systems from their vendors such as ISO 17025. Originally known as ISO/IEC Guide 25, ISO 17025 adds the concept of competence and traceability into the equation. It applies directly to those organisations that produce testing and calibration results. Part of the requirement of ISO 17025 is reporting UoM.

Ideally a particle counter calibration laboratory that calibrates to ISO 21501 and is accredited to ISO 17025, NIST, NATA, UKAS etc, is the best option. At the moment not many labs with this type of accreditation exist but this is likely to change, based on the emergence of tighter controls and higher expectations from customers using critical monitoring equipment such as particle counters.

UoM, as mentioned, gives the user something to work with and, considering the importance of particle counters in today’s industries, the extra confidence given with a UoM report significantly helps the decisions some Quality Managers and Qualified Persons face on a daily basis in the pharmaceutical sectors.

Manufacturers’ response

Some particle counter manufacturers have already begun to pay attention, with new models on the market stating compliance to ISO 21501, thus illustrating their agreement and willingness to work with the new standard.

One particular manufacturer has risen to the challenge leading the way in validating its worldwide service centres by using the same calibration bench setup with reference equipment traceable to the gold standard CPC reference unit called the “Universal Reference Unit”. This approach has increased the repeatability and accuracy of calibrations performed outside of the factory. All calibration technicians must be factory trained and each service centre is revalidated periodically to ensure the high standards are kept in place. The calibration procedure itself reduces errors in reference particle aerosol generation and is automated to eliminate the human error factor taking advantage of electronic adjustments during the calibration process.

ISO 21501-4 states: “Instruments that conform to this part of ISO 21501 are used for the classification of air cleanliness in cleanrooms and associated controlled environments in accordance with ISO 14644-1”. The next revision of ISO 14644-1 refers to ISO 21501-4 “A calibration standard for air particle counters ratified in 2007”.

The accuracy of a particle counter is an important tool in the fight against cleanroom contamination and ISO 21501 means a more accurate particle counter calibration.

Summary of ISO 21501 requirements

- Size calibration • Verification of size setting • Counting efficiency • Size resolution • False count rate; maximum particle number concentration • Sampling flow rate • Sampling time • Sampling volume for liquid particle counters • Calibration interval to be defined

Contact Jason Kelly Calibration Services Engineer Kenelec Scientific T **+61 (03) 98731022** "jason@kenelec.com.au"

